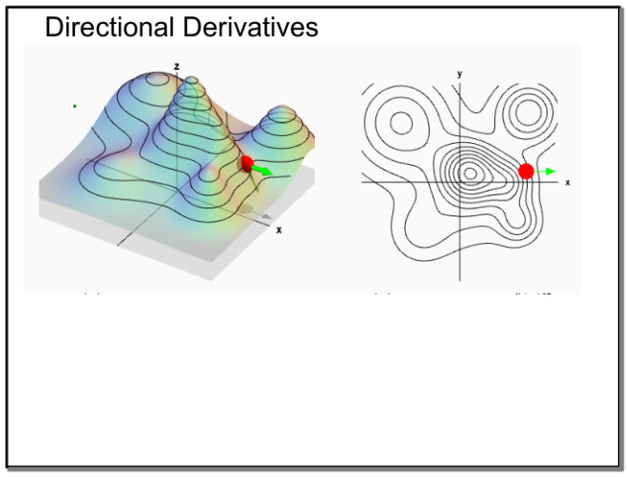
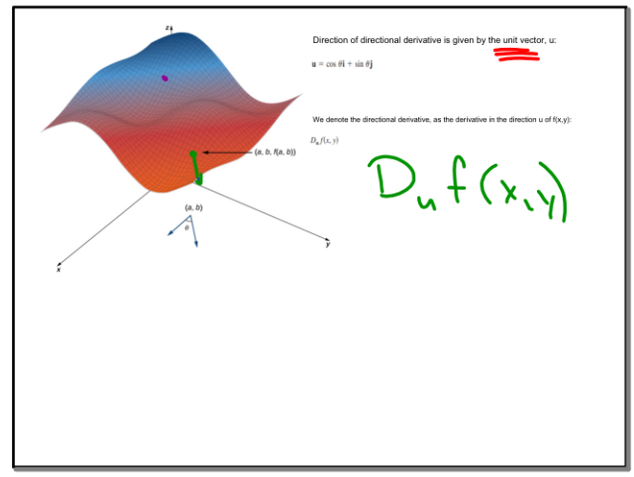
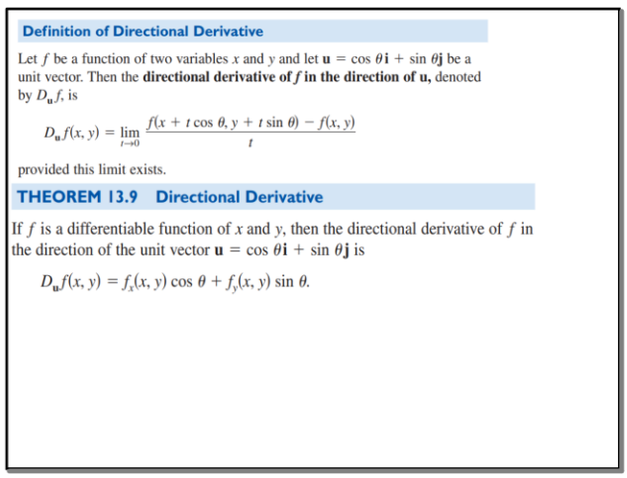


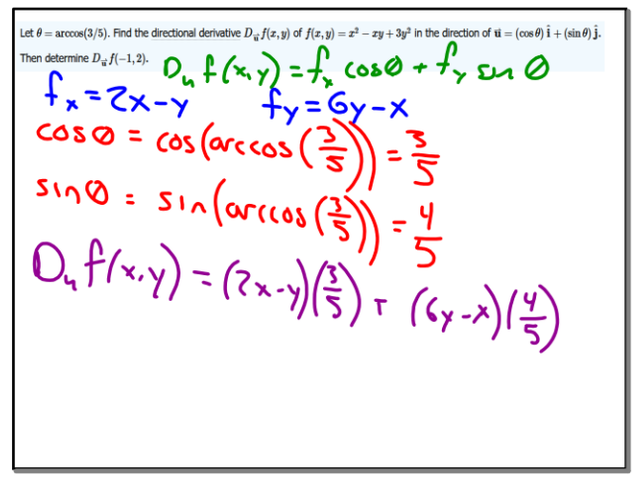
Directional derivatives: f_x and f_y only give the slope in the x and y directions. The directional derivative gives the slope of the surface in any direction you choose, like the arrow on the contour map.



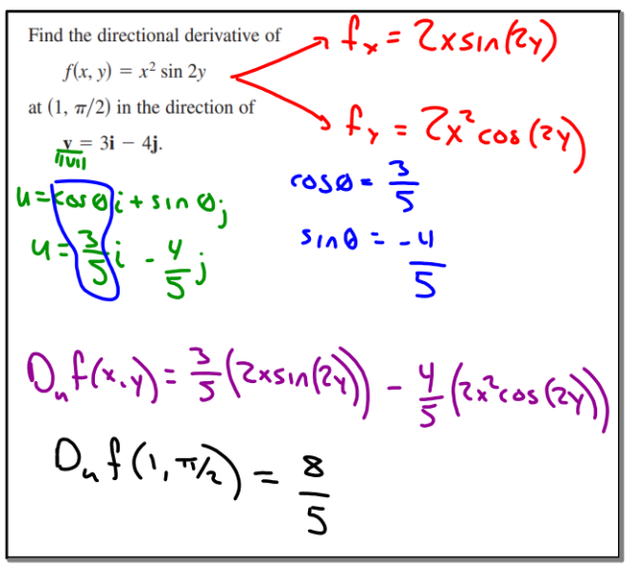
The direction is a unit vector u = cos θ i + sin θ j. D_u f(x, y) is the rate of change of f as you move from (x, y) in the direction u.



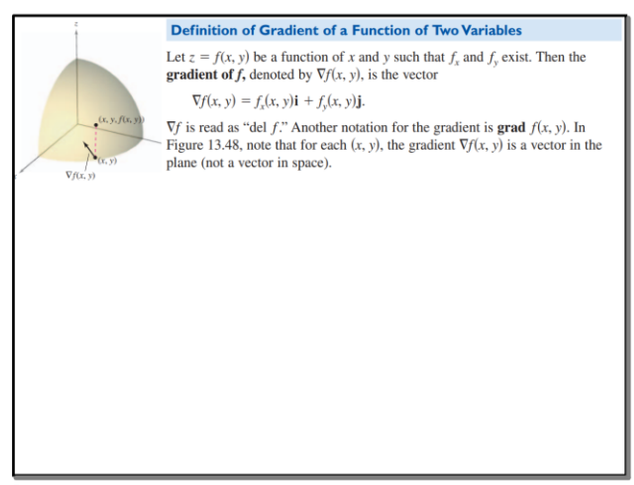
Definition: D_u f is the limit of [f(x + t cos θ, y + t sin θ) − f(x, y)]/t as t → 0. In practice use Theorem 13.9: D_u f(x, y) = f_x cos θ + f_y sin θ.



Process: find f_x = 2x − y and f_y = 6y − x, get cos θ = 3/5 and sin θ = 4/5 from θ = arccos(3/5), then D_u f = (3/5)(2x − y) + (4/5)(6y − x). At (−1, 2): (3/5)(−4) + (4/5)(13) = 8.



If the direction is given as a vector, divide by its length first: v = 3i − 4j has length 5, so u = (3/5)i − (4/5)j. At (1, π/2), f_x = 2x sin 2y = 0 and f_y = 2x² cos 2y = −2, so D_u f = 0 + (−2)(−4/5) = 8/5.



The gradient packs both partials into one vector: ∇f(x, y) = f_x i + f_y j, read "del f". It lives in the xy-plane, not in space.

Find the gradient of $f(x, y) = y \ln x + xy^2$ at the point $(1, 2)$.

$f_x = \frac{y}{x} + y^2$

$f_y = \ln x + 2xy$

$\nabla f(x,y) = \left(\frac{y}{x} + y^2\right)i + (\ln x + 2xy)j$

$\nabla f(1,2) = 6i + 4j$

Find each partial, then plug in. f = y ln x + xy²: f_x = y/x + y² and f_y = ln x + 2xy, so ∇f = (y/x + y²)i + (ln x + 2xy)j and ∇f(1, 2) = 6i + 4j.

THEOREM 13.10 Alternative Form of the Directional Derivative

If f is a differentiable function of x and y, then the directional derivative of f in the direction of the unit vector $\mathbf{u}$ is

$D_{\mathbf{u}} f(x, y) = \nabla f(x, y) \cdot \mathbf{u}$.

$(f_x i + f_y j) \cdot (\cos\theta\, i + \sin\theta\, j)$

$f_x \cos\theta + f_y \sin\theta$

Theorem 13.10: the directional derivative is a dot product, D_u f = ∇f · u = (f_x i + f_y j)·(cos θ i + sin θ j) = f_x cos θ + f_y sin θ. Find ∇f once, then dot it with any unit direction.

Find the directional derivative of

$f(x, y) = 3x^2 - 2y^2$

at $\left(-\frac{3}{4}, 0\right)$ in the direction from $P\left(-\frac{3}{4}, 0\right)$ to $Q(0, 1)$.

$PQ = \langle \frac{3}{4}, 1 \rangle$

$f_x = 6x$

$f_y = -4y$

$\|PQ\| = \sqrt{\frac{25}{16}} = \frac{5}{4}$

$\frac{PQ}{\|PQ\|} = \langle \frac{3}{5}, \frac{4}{5} \rangle$

$D_u f(x,y) = (6x)\left(\frac{3}{5}\right) + (-4y)\left(\frac{4}{5}\right)$

$\cos\theta = \frac{3}{5}$

$D_u f\left(-\frac{3}{4}, 0\right) = (6)\left(-\frac{3}{4}\right)\left(\frac{3}{5}\right) = -\frac{27}{10}$

$\sin\theta = \frac{4}{5}$

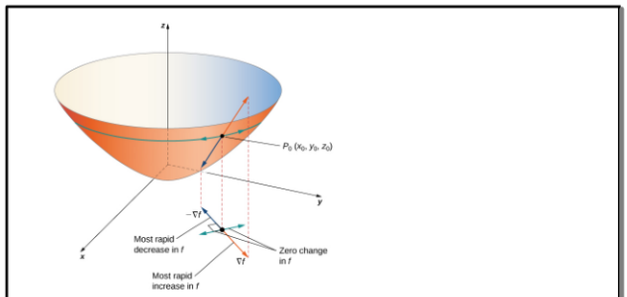
Direction from P to Q: PQ = ⟨3/4, 1⟩ with length 5/4, so u = ⟨3/5, 4/5⟩. With f_x = 6x and f_y = −4y, D_u f = (3/5)(6x) + (4/5)(−4y). At (−3/4, 0): (3/5)(−9/2) = −27/10.

THEOREM 13.11 Properties of the Gradient

Let f be differentiable at the point (x, y).

1. If $\nabla f(x, y) = \mathbf{0}$, then $D_{\mathbf{u}} f(x, y) = 0$ for all $\mathbf{u}$.
2. The direction of *maximum* increase of f is given by $\nabla f(x, y)$. The maximum value of $D_{\mathbf{u}} f(x, y)$ is $\|\nabla f(x, y)\|$.
3. The direction of *minimum* increase of f is given by $-\nabla f(x, y)$. The minimum value of $D_{\mathbf{u}} f(x, y)$ is $-\|\nabla f(x, y)\|$.

Theorem 13.11: ∇f points the way f increases fastest, at rate ‖∇f‖. −∇f is the way of fastest decrease, at rate −‖∇f‖. If ∇f = 0, every directional derivative is 0.



On the surface: the gradient direction is steepest uphill, the opposite direction is steepest downhill, and directions perpendicular to ∇f give zero change (they follow the level curve).

The temperature in degrees Celsius on the surface of a metal plate is

$T(x, y) = 20 - 4x^2 - y^2$

where x and y are measured in centimeters. In what direction from $(2, -3)$ does the temperature increase most rapidly? What is this rate of increase?

$\nabla T = -8x\, i - 2y\, j$

$\nabla T(2,-3) = -16i + 6j$

$\|\nabla T(2,-3)\| = 17.088 = 2\sqrt{73}$

∇T = −8x i − 2y j, so ∇T(2, −3) = −16i + 6j: the direction of fastest temperature increase. The rate is ‖∇T‖ = √(256 + 36) = √292 = 2√73 ≈ 17.09 °C per cm.

A heat-seeking particle is located at the point $(2, -3)$ on a metal plate whose temperature at (x, y) is

$$T(x, y) = 20 - 4x^2 - y^2.$$

Find the path of the particle as it continuously moves in the direction of maximum temperature increase.

Not worked on the slide. The particle always moves along $\nabla T = \langle -8x, -2y \rangle$, so $dx/dt = -8x$ and $dy/dt = -2y$. Starting from $(2, -3)$: $x = 2e^{(-8t)}$ and $y = -3e^{(-2t)}$, so the path is $x = 2y^4/81$.

THEOREM 13.12 Gradient Is Normal to Level Curves

If f is differentiable at (x_0, y_0) and $\nabla f(x_0, y_0) \neq \mathbf{0}$, then $\nabla f(x_0, y_0)$ is normal to the level curve through (x_0, y_0).

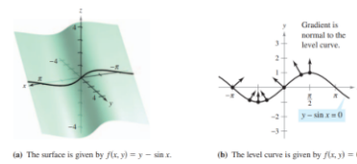
Theorem 13.12: if $\nabla f(x_0, y_0) \neq 0$, the gradient is perpendicular (normal) to the level curve through (x_0, y_0). That is why it points straight uphill, across the contours.

Sketch the level curve corresponding to $c = 0$ for the function given by

$$f(x, y) = y - \sin x$$

and find a normal vector at several points on the curve.

Not worked on the slide. The $c = 0$ level curve of $f = y - \sin x$ is $y = \sin x$. $\nabla f = -\cos x\, i + j$ is normal to it: for example $\langle -1, 1 \rangle$ at $(0, 0)$, $\langle 0, 1 \rangle$ at $(\pi/2, 1)$ and $\langle 1, 1 \rangle$ at $(\pi, 0)$.



(a) The surface is given by $f(x, y) = y - \sin x$. (b) The level curve is given by $f(x, y) = 0$.

The picture for the previous problem: the surface $z = y - \sin x$ on the left, and on the right its level curve $y = \sin x$ with gradient arrows drawn perpendicular to it.

Find $\nabla f(x, y, z)$ for the function given by

$$f(x, y, z) = x^2 + y^2 - 4z$$

and find the direction of maximum increase of f at the point $(2, -1, 1)$.

Not worked on the slide. Gradients work the same way in 3D: $\nabla f = 2x\, i + 2y\, j - 4k$. At $(2, -1, 1)$, $\nabla f = 4i - 2j - 4k$ is the direction of maximum increase, at rate $\|\nabla f\| = \sqrt{36} = 6$.